

BIJAY SHAKYA

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Google Scholar](#)

PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Science with expertise in computer vision, deep learning, object detection, and federated learning. Proven experience optimizing and benchmarking lightweight models, deploying ONNX and TensorRT inference pipelines on NVIDIA Jetson Orin Nano, and conducting federated learning simulations with Flower and Docker. Published researcher with work in IEEE Access, CVPR Workshops, HICSS, and CAC. Demonstrated ability to bridge advanced research and practical engineering, delivering efficient AI solutions for real-world applications.

EDUCATION

Ph.D. in Computer Science Dakota State University, Madison, SD <i>Research Areas: Deep Learning, Computer Vision, Object Detection, Federated Learning</i>	August 2023 – December 2026 (Expected)
Master of Science in Computer Science Dakota State University, Madison, SD	January 2019 – December 2019
Bachelor of Engineering in Computer Engineering Kathmandu University, Nepal	August 2011 – June 2015

TECHNICAL SKILLS

Research Areas: Computer Vision, Object Detection, Remote Sensing, Hyperspectral Imaging, Vision-Language Models (VLMs), Super-Resolution, Transfer Learning, Fine-tuning, Federated Learning
Programming & Software Development: Python (3 years), C# (6 years), C, OOP, SQL, JavaScript, HTML5, CSS3
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Flower Framework
Deployment & Edge: Docker, Azure CI/CD pipelines, TensorRT, ONNX, Jetson Orin Nano, Linux
Backend & Web: Blazor, .NET Core 2.1, ASP.NET MVC, REST APIs, Entity Framework Core, LINQ, Web Services, SQL Server, MySQL, PostgreSQL, AWS (EC2, S3, VPC, CloudWatch), Angular, DevExpress, Kendo UI
Testing & Tools: Git, GitHub, JupyterLab, NUnit, MSTest, Selenium, Visual Studio, Cursor, Copilot, SVN, TFS

AI/ML RESEARCH & ACADEMIC EXPERIENCE

Graduate Research Assistant – Dakota State University, Madison, SD Lightweight Object Detection & Edge AI: Benchmarked and fine-tuned eight lightweight YOLO models on VisDrone2019, DOTA-v1.5, and xView for cross-platform, real-time small-object detection using a Jetson Orin Nano and an A100 GPU-accelerated HPC cluster. On Jetson Orin Nano, the TensorRT FP16 model achieved over 100 FPS with 10 ms latency and a 3–5× speedup over the PyTorch version. IEEE Access paper Multimodal Remote Sensing AI: Designed and evaluated a post-disaster structural damage assessment pipeline on a shared GPU-accelerated HPC cluster, integrating super-resolution, YOLOv11 object detection, and vision-language models on xBD subsets (Moore Tornado and Hurricane Matthew). On the Moore Tornado subset, Qwen3-VL-32B achieved 87.1% accuracy and 0.873 F1-score in distinguishing low-risk from high-risk structural damage. CVPR Workshops paper · Code Secure AI for Hyperspectral Imaging: Engineered a federated AI-driven crop anomaly detection framework using hyperspectral imagery to detect early- and mid-stage crop defects while analyzing privacy–utility tradeoffs in distributed learning. Accepted at CAC 2026. Federated Learning & ML Deployment: Built a Dockerized federated learning robustness testbed on a Linux VM using Python, PyTorch, and Flower to orchestrate distributed client-server training, simulate poisoning attacks, and evaluate robust aggregation defenses under reproducible ML experimentation settings. Efficient Small-Object Detection: Developed a resource-efficient, density-aware multi-level slicing inference pipeline for small-object detection, reducing inference time by 93–98% on VisDrone2019 and by 78% on xView compared to baseline SAHI method, while maintaining detection accuracy. Research & Teaching: Authored peer-reviewed research in IEEE Access, CAC, and CVPRW venues; served as graduate teaching assistant for a graduate-level Research and Design Methods course.	August 2024 – Present
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Student Researcher – Dakota State University, Madison, SD	August 2023 – July 2024
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- Improved small-object detection by ensembling YOLO variants with Weighted Boxes Fusion: **+5% mAP@0.5** over the best single YOLO baseline.
- Researched quantum computing security implications for Smart Grid systems and analyzed emerging threats with corresponding mitigation strategies.

Research Assistant – Dakota State University

January 2019 – December 2019

- Delivered technical training workshops on cloud computing using AWS EC2 and S3 to **20+ participants** with hands-on exercises and training materials.
- Coordinated undergraduate students in Internet of Things security research during the Research Experience for Undergraduates summer program.
- Designed a web application using Laravel, MySQL, and AWS APIs to provide virtual computing lab services for university students.

PROFESSIONAL SOFTWARE ENGINEERING EXPERIENCE

Software Engineer – United States

February 2020 – December 2022

Ferrandino & Son, Audubon, PA | IMCS Group, Irving, TX

- Deployed an enterprise application in C#, Blazor, and SQL Server that supported contractor-assignment workflows across construction business units, and improved REST API performance by **40%**.
- Architected a retail delivery and delivery-cost system covering FootLocker stores in the USA, Canada, and Europe, and reduced manual store-location updates by **80%**.
- Reduced manual reporting by **10+ hours per week** with an automated reporting system in C# and ITextSharp.
- Integrated Azure CI/CD pipelines to automate build, testing, and continuous deployment of application features and enhancements, improving release consistency and reducing manual deployment effort.
- Improved reliability of enterprise data operations with SQL triggers, views, stored procedures, and automation scripts for data manipulation workflows.

Software Engineer – Nepal

April 2016 – December 2018

Rigo Technologies | Spiralogics Inc, Kathmandu

- Led a **backend team of five members** that shipped a tenant-based e-learning platform in C#, .NET Core, PostgreSQL, and Entity Framework Core.
- Launched real-time webhook integrations for video analytics in production, with error handling and monitoring for system reliability.
- Reduced workload by **80%** with an ETL module that validated data and sent email alerts, and cut CRUD maintenance by moving complex SQL into stored procedures, views, and LINQ.
- Developed a workers' compensation and healthcare system connecting injured parties with healthcare providers for medications and managing billing reports.

Software Engineer Intern – Nepal

September 2015 – April 2016

Verisk Health, Kathmandu

- Improved healthcare document parsing speed by **25%** with a C# and XML report parser, and integrated module into a web application.
- Developed healthcare application modules in C#/ASP.NET MVC, with unit tests in NUnit and automated UI regression tests in Selenium.
- Maintained and enhanced an ASP.NET Web Forms healthcare application, delivering feature updates and bug fixes.

SELECTED PUBLICATIONS

- **B. Shakya**, C. Hoier, and K. M. Ahmed. [From Pixels to Semantics: A Multi-Stage AI Framework for Structural Damage Detection in Satellite Imagery](#). *CVPR Workshop*, 2026.
- **B. Shakya**, O. El-Gayar, J. Kaabi, and K. M. Ahmed. [Small-Object Detection at the Edge: A Pareto-Efficient Benchmark of Lightweight YOLO Models on UAV and Overhead Datasets](#). *IEEE Access*, 2025.
- Y. Wang, K.N. Muthusamy Ragotheraman, **B. Shakya**. [Towards Trusted Data Processing for Information and Intelligence Systems](#). *Hawaii International Conference on System Sciences (HICSS)*, 2021.

AWARDS & CERTIFICATIONS

- Graduate Research Initiative Award, Dakota State University.
- Second Place, Third Annual Data Analytics Competition, Dakota State University.

August 2023
December 2024